

ESN–A2 Hyperparameter Search: Full-Parameter Grid with Smooth Surrogate Statistics, Optimised via Bounded Levenberg–Marquardt

Protocol note for the ESN A2-981003 pipeline, market-data variant

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Abstract

This note determines the optimal ESN–A2 architecture **hyperparameters** (reservoir sizes N_r, N_z and 7 shared z-bank hyperparameters) against real S&P 500 market data using **bounded Levenberg–Marquardt** (`scipy.optimize.least_squares, method="trf"`) as the minimisation algorithm – not a grid. **Every one of the 12 entries of the calibrated parameter vector p is gridded**: a full $2^{12} = 4,096$ -point factorial is computationally intractable inside every hyperparameter evaluation, so a randomised **2-level fractional design** (24 points) is used instead, over a wide admissible range for each of the 12 entries. At every grid point the ESN is simulated and its 11 stylised facts computed using **smooth surrogate statistics**; these are averaged across the grid into one representative vector, compared against each of 5 real assets to build a 55-entry weighted residual vector, whose sum of squares bounded LM/TRF minimises over the 9 architecture hyperparameters. Results are reported on the **Gaussian-kernel score** $S_{\text{avg}} \in [0, 15.3]$ (0 = perfect match, lower is better).

Bounded LM’s best point converges to $N_r = 22, N_z = 29$: S_{avg} ’s running minimum falls from 2.90 (start) to 2.62 over 200 iterations, though this 2.62 is a single noisy Monte-Carlo evaluation, not the architecture’s true score – repeated evaluation gives $S_{\text{avg}} = 2.725 \pm 0.043$ (5 seeds), matched by the (N_r, N_z) heat map’s best evaluated cell, $N_r = 22, N_z = 40$, at 2.708. These repeated-evaluation figures (2.71–2.73), not the single-draw 2.62, are this note’s reliable estimate of the architecture’s performance.

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1 The model

1.1 Two parameter classes – hyperparameters vs. parameters

The ESN–A2 pipeline (`esn_base.py`) separates two conceptually distinct sets of quantities:

- **Hyperparameters** = $\{N_r, N_z\} \cup \{\text{z_strength}, \text{even_strength}, \text{linear_strength}, \text{gamma_norm}, \text{local_z_strength}, \text{zz_scale}, \text{sign_prob_neg}\}$: 9 values defining the architecture, shared across every calibration target. $N_r \in [8, 200]$, $N_z \in [2, 60]$ (log-encoded, rounded to the nearest integer); the 7 shared hyperparameters have admissible ranges fixed in the code (`HP_SPEC`):

Hyperparameter	lo	hi
<code>z_strength</code>	0.05	2.0
<code>even_strength</code>	0.10	5.0
<code>linear_strength</code>	0.01	2.0
<code>gamma_norm</code>	0.10	3.0
<code>local_z_strength</code>	0.001	0.5
<code>zz_scale</code>	0.01	0.5
<code>sign_prob_neg</code>	0.05	0.50

These 9 quantities are determined in this note by bounded Levenberg–Marquardt, not a grid.

- **Parameters** $p = (H, \lambda_{lo}, \lambda_{hi}, a_{lo}, a_{hi}, \Delta b_0, \text{scale}, \text{rough_scale}, \text{zr_lo}, \text{zr_hi}, m_1, m_2)$: 12 values ordinarily calibrated per scenario. **This note grids all 12** (§1.6).

1.2 ESN architecture

Two coupled layers, both driven by a single shared innovation $\varepsilon_t \sim \mathcal{N}(0, 1)$ per step, updated once per day ($dt = 1$).

The reservoir. $r = (r_1, \dots, r_{N_r})$ is a bank of independent Ornstein–Uhlenbeck processes with geometrically spaced rates

$$\lambda_i = \lambda_{lo} \left(\frac{\lambda_{hi}}{\lambda_{lo}} \right)^{\frac{i-1}{N_r-1}}, \quad i = 1, \dots, N_r, \quad (1)$$

so $\lambda_1 = \lambda_{lo}$, $\lambda_{N_r} = \lambda_{hi}$, and every intermediate rate is log-uniformly spaced between them – all N_r modes driven by the *same* ε_t (not N_r independent shocks):

$$r_{t+1,i} = \alpha_i r_{t,i} + c_i \varepsilon_t, \quad \alpha_i = e^{-\lambda_i dt}, \quad c_i = \sqrt{1 - \alpha_i^2}, \quad (2)$$

so each mode’s own memory is set purely by its own λ_i : fast modes (λ_i large) have $\alpha_i \approx 0$ and retain almost none of their previous value each step; slow modes (λ_i small) have $\alpha_i \approx 1$ and evolve gradually.

The z -bank. $z = (z_1, \dots, z_{N_z})$ is a nonlinear bank with its own geometrically spaced decay rates, by the same construction as Eq. 1,

$$a_j = a_{lo} \left(\frac{a_{hi}}{a_{lo}} \right)^{\frac{j-1}{N_z-1}}, \quad j = 1, \dots, N_z, \quad (3)$$

each unit reading a fixed fast/slow pair of reservoir taps, $r_{\text{fast}(j)}$ and $r_{\text{slow}(j)}$ (2 fixed indices into r per z -bank unit, spanning the reservoir's own range of modes). Writing $p_1 = r_{t+1, \text{fast}(j)}$, $p_2 = r_{t+1, \text{slow}(j)}$ (the reservoir state *after* the Eq. 2 update just applied), each unit updates as

$$z_{t+1,j} = e^{-a_j dt} z_{t,j} + (1 - e^{-a_j dt}) m_{t+1} \mathbf{z_strength} \tanh(u_{t,j}), \quad (4)$$

where the driving signal $u_{t,j}$ combines a cross term, an even (quadratic) term, a linear term, and a local-neighbour coupling to the z -bank's own adjacent units,

$$u_{t,j} = s_j p_1 p_2 + \mathbf{even_strength} (0.7 p_1^2 + 0.3 p_2^2 - 1) - \mathbf{linear_strength} (0.7 p_1 + 0.3 p_2) + \zeta_j z_{t,j} + \frac{1}{2} \mathbf{local_z_strength} (z_{t,j-1} + z_{t,j+1}), \quad (5)$$

with s_j, ζ_j fixed per-unit sign/scale constants set once when the matrices are built, and $j \pm 1$ clipped to the bank's own boundary. The scalar gate

$$m_{t+1} = \max \left(0, 1 - \mathbf{gamma_norm} \frac{\|r_{t+1}\|}{\sqrt{N_r}} \right) \quad (6)$$

suppresses the z -bank's own update whenever the reservoir's current norm is large, so the 2 layers do not reinforce each other's own extremes unchecked.

Pre-activation and return. Using r_t, z_t (the state *before* Eqs. 2–4's own update this step),

$$\eta_t = b_0 + \rho_r \mathbf{rough_scale} (q^\top r_t) + \sum_{j=1}^{N_z} w_{j,z} z_{j,t} + r_t^\top Q r_t, \quad \sigma_t = \sqrt{\mathbf{SIG_MIN}^2 + \mathbf{softplus}(\eta_t)^2},$$

with $w_{j,z} = \mathbf{z_readout}_j / \sqrt{N_z}$ geometrically profiled via $(\mathbf{zr_lo}, \mathbf{zr_hi})$, and $Q = \frac{1}{N_r} (m_1 I_{N_r} + m_2 q q^\top)$. Daily log-returns are $dX_t = (\mu - \frac{1}{2} \sigma_t^2) + \sigma_t \varepsilon_t$, reusing the *same* ε_t that Eq. 2 then uses to update r_{t+1} – the return and the reservoir's own next state are never independent draws.

1.3 Data source: 5 real market assets

Five randomly selected tickers from the S&P 500 panel (`list_dics_SP500_ohlcvol_19940101_filtered.pkl`, 243 tickers, daily OHLCV, 1994–2025), each used over its *entire* available history: GPC, QCOM, INTC, LUV, NI. Each asset's reference is built by splitting its full history into 6 non-overlapping time windows, computing the 11 (smooth-surrogate, §1.4) stylised facts once per window, and feeding that list of 6 stat dicts to `make_score_ref`. Variance proxy: Garman–Klass, $v_t = \frac{1}{2} (\ln(H_t/L_t))^2 - (2 \ln 2 - 1) (\ln(C_t/O_t))^2$, paired with $x_t = \ln(C_t/C_{t-1})$.

1.4 Smooth surrogate statistics

The 11 stylised facts are not all equally smooth (differentiable, low-variance) functions of a simulated path. Several are built from a HARD order statistic or indicator: piecewise-constant quantities whose value jumps discretely as the underlying data changes, rather than responding continuously

– these jumps inject avoidable noise and zero-gradient regions into the optimisation landscape, on top of whatever smoothness the score kernel itself has (§1.5). **6 of the 11 are already smooth** (means or mean correlations) and are used exactly as defined in the base pipeline; **5 are hard**, and each is replaced here by a smooth surrogate that recovers the original definition in a limit, while remaining a genuinely differentiable function of the data everywhere.

Notation: $sd_t = \sqrt{v_t}$ (daily vol from the Garman–Klass proxy v_t), x_t the daily log-return, $\text{ann} = \sqrt{252}$; $\text{abs_acf}(L)$, $\text{sq_acf}(L)$, $\text{ret_acf}(L)$ the lag- L autocorrelations of $|x_t|$, x_t^2 , and x_t respectively, for $L = 1, \dots, 20$.

- **$\hat{H}$ – log-spaced structure-function regression.** The original estimator regresses $\log(\text{structure function})$ on $\log(\text{lag})$ using only 7 *fixed* dyadic lags $\{1, 2, 4, 8, 16, 32, 64\}$ (filtered by a hard cutoff $L < n/4$) – a sparse, arbitrary design whose own lag *selection* is itself a hard, non-smooth choice. The surrogate keeps the same log-uniform lag *spacing* (a naive dense/linear lag grid was tested and found to shift the estimate’s mean without reducing its variance – log-uniform spacing avoids over-weighting short lags) but uses $n_{\text{lags}} = 15$ points spanning the *full* admissible range $[1, n/4]$ instead of 7 fixed points capped at 64:

$$d(L) = \overline{(\log v_{t+L} - \log v_t)^2}, \quad \hat{H}_{\text{smooth}} = \frac{1}{2} \text{slope}_L(\log L, \log d(L)), \quad L \in \{L_1, \dots, L_{15}\} \text{ log-spaced in } [1, n/4]$$

- **99.5% quantile of ann. vol – Harrell–Davis estimator.** The hard version returns a single order statistic, $sd_{(\lceil 0.995n \rceil)}$: a step function of the data that jumps whenever the ranking changes. The surrogate is a *Beta-weighted average of every order statistic*: with $sd_{(1)} \leq \dots \leq sd_{(n)}$ the sorted daily vols, $q = 0.995$, $a = (n+1)q$, $b = (n+1)(1-q)$, and $I_z(a, b)$ the regularised incomplete Beta function (the CDF of $\text{Beta}(a, b)$ at z):

$$w_i = I_{i/n}(a, b) - I_{(i-1)/n}(a, b), \quad q995_{\text{smooth}} = \text{ann} \cdot \sum_{i=1}^n w_i sd_{(i)}.$$

The weights w_i are concentrated near rank qn but decay smoothly, so a small change in the data smoothly reweights the sum instead of discretely swapping which single point is reported.

- **Max ann. vol, max|return ACF| – log-sum-exp soft-max.** The hard max is non-differentiable at ties and has zero gradient with respect to every point except the single maximiser. The surrogate is the standard soft-max: with $\beta = 8/\hat{\sigma}_x$ ($\hat{\sigma}_x$ the sample std of the values being maximised over) and $m = \max_t x_t$,

$$\text{softmax}(x) = m + \frac{1}{\beta} \log \left(\frac{1}{T} \sum_{t=1}^T e^{\beta(x_t - m)} \right) \xrightarrow{\beta \rightarrow \infty} \max_t x_t,$$

applied to $\{sd_t\}$ (then scaled by ann) for max ann. vol, and to $\{|\text{ret_acf}(L)|\}_{L=1}^{20}$ (with its own β) for max|return ACF|. Every point contributes a (small but nonzero) gradient, not just the maximiser.

- **Taylor fraction – sigmoid relaxation.** The hard version is $\frac{1}{20} \sum_L \mathbb{K}[\text{abs_acf}(L) > \text{sq_acf}(L)]$: an indicator that is exactly 0 or 1 per lag, with a zero-gradient plateau on both sides of the crossing. Replacing $\mathbb{K}[\cdot > 0]$ with the logistic sigmoid $\sigma(z) = 1/(1 + e^{-z})$, $\kappa = 200$:

$$\text{taylor_frac}_{\text{smooth}} = \frac{1}{20} \sum_{L=1}^{20} \sigma(\kappa [\text{abs_acf}(L) - \text{sq_acf}(L)]) \xrightarrow{\kappa \rightarrow \infty} \text{the hard version exactly},$$

giving a smooth, non-zero gradient in a neighbourhood of every crossing rather than an exact step.

- **Excess kurtosis – tanh-winsorised 4th moment.** The hard 4th moment, $\overline{c_t^4}/\text{var}^2 - 3$ with $c_t = x_t - \bar{x}$, is notoriously high-variance: a single extreme return can dominate the whole statistic because it enters raised to the 4th power. The surrogate soft-clips c_t *before* the 4th power, via $c_t^{\text{soft}} = c_s \tanh(c_t/c_s)$ with clip scale $c_s = 12\sqrt{\text{Var}(x)}$ (chosen, by direct comparison over 20 simulated paths, to cut run-to-run standard deviation by roughly 40% while leaving the kurtosis level recognisably similar to the hard version's):

$$\text{kurt}_{\text{smooth}} = \frac{(\overline{c^{\text{soft}}})^4}{\text{var}^2} - 3.$$

$\tanh$ is C^∞ , so this remains a smooth function of every return, including the extreme ones, instead of letting one outlier dominate.

Mean ann. vol ($\text{ann} \cdot \overline{sd_t}$), mean vol ACF ($\frac{1}{20} \sum_L \text{corr}(v_t, v_{t+L})$), Taylor gap ($\overline{\text{abs_acf}} - \overline{\text{sq_acf}}$), Zumbach asymmetry (the past-return²/future-vol vs. past-vol/future-return² correlation difference, averaged over horizons $L \in \{5, 10, 20\}$), and leverage correlation ($\frac{1}{20} \sum_L \text{corr}(x_t, v_{t+L})$) are all already means or mean correlations – smooth, low-variance functions of the path – and are used exactly as defined in the base pipeline, with no surrogate needed.

1.5 Score function

Every one of the 11 stylised facts x_k is compared to a target centre c_k and tolerance s_k (both derived per market asset, §1.3, from the spread of that statistic across the asset's own 6 time windows) via the normalised deviation $z_k = (x_k - c_k)/s_k$. Results in this note are reported on the **Gaussian-kernel score** S_{avg} :

$$d_k = w_k \left(1 - e^{-\frac{1}{2}z_k^2}\right) \in [0, w_k], \quad S_{\text{scenario}} = \sum_{k=1}^{11} d_k + 5 \cdot \text{Stress}, \quad S_{\text{avg}} = \frac{1}{5} \sum_{\text{asset}=1}^5 S_{\text{scenario}}^{(\text{asset})},$$

bounded in $[0, 15.3]$ per scenario (the sum of the 11 weights, 10.3, plus a 5-point penalty, **Stress**, that fires if the simulated path's volatility is extreme or implausibly small/large). $S_{\text{avg}} = 0$ is a perfect match on every term for every asset; $S_{\text{avg}} = 15.3$ is the worst possible score.

Internally, bounded LM (`scipy.optimize.least_squares`) minimises the sum of squares of the residual vector $r_k^{(\text{asset})} = \sqrt{w_k} z_k^{(\text{asset})}$ (one-sided terms clipped at 0), concatenated across all 5 assets into a 55-entry vector (§1.6) – this is what a least-squares solver requires as its objective, and its minimum tracks S_{avg} 's minimum closely since both are increasing functions of the same normalised deviations z_k . All figures, tables, and reported numbers in this note use S_{avg} .

Weight w_k for each of the 11 terms: $\hat{H}$ (2.2), mean ann. vol (1.4), 99.5% quantile ann. vol (0.6), max ann. vol (0.2), mean vol ACF (0.8), Taylor gap (1.0), Taylor fraction (0.8), Zumbach asymmetry (1.0), leverage corr. (1.1), excess kurtosis (0.7), max|return ACF| (0.5) – summing to 10.3. $\hat{H}$ and leverage correlation carry the largest weights (matching, and most directly tied to the persistence and asymmetry properties this architecture family was originally designed to reproduce); the two tail-volatility terms (99.5% quantile, max) carry the smallest, since they are the most estimation-noisy even after smoothing (§1.4).

1.6 The procedure: full- p -grid, smooth surrogates, hyperparameters via bounded LM

Step 1 – grid ALL 12 entries of p , over a wide admissible range.

Entry	low	high	Entry	low	high
H	0.01	0.3	zr_lo	0.01	0.1
λ_{lo}	10^{-5}	10^{-1}	zr_hi	0.1	0.4
λ_{hi}	1.0	5.0	m_1	-0.1	0.1
rough_scale	0.2	0.7	m_2	-0.1	0.1
a_{lo}	$\frac{1}{450}$	$\frac{1}{350}$	Δb_0	-0.02	0.02
a_{hi}	$\frac{1}{9}$	$\frac{1}{6}$	scale	0.98	1.02

A randomised 2-level fractional design of $N = 24$ points is used (each point independently sets each of the 12 entries to its low or high value, fixed seed): a full $2^{12} = 4,096$ -point factorial remains intractable, but every entry is still gridded (appears at both levels across the design).

Step 2 – average the smooth stylised facts over that grid. For a given hyperparameter candidate: at each of the 24 grid points, simulate $N_{\text{CAL_PATHS}} = 10$ ESN paths, each $T_{\text{CAL}} = 1342$ days long – matched exactly to the market data’s own scale: all 5 assets share an identical history length (8,054 trading days), split into 6 non-overlapping windows of $8054/6 = 1342$ days each to build the target references (§1.3), so a simulated path is now the same length as the real windows it is compared against, rather than an arbitrary shorter horizon. Compute the 11 smooth stylised facts, average across the 10 paths, then **average across the 24 grid points** – never a minimum – into one representative vector.

Step 3 – build a 55-entry weighted residual vector. Compare that vector to each of the 5 assets’ own references:

$$r_k^{(\text{asset})} = \sqrt{w_k} \frac{x_k - c_k^{(\text{asset})}}{s_k^{(\text{asset})}}, \quad k = 1, \dots, 11, \quad \text{asset} = 1, \dots, 5,$$

(one-sided terms clipped at 0), giving 55 residuals – a well-posed least-squares problem against the 9 hyperparameters.

Step 4 – minimise with bounded Levenberg–Marquardt. Bounded LM (`method="trf"`) minimises $\sum (r_k^{(\text{asset})})^2$ over the 9-dimensional hyperparameter vector, native box bounds enforced directly. Initialised at the previously reported A2-981003 hyperparameters and $N_r = 48, N_z = 20$; `diff_step=0.1`.

1.7 Algorithm (pseudocode)

2 Implementation notes

1. **Aggregate (all-5-assets) residuals keep the system well-posed.** An 11-residual, per-asset version of this procedure is underdetermined against 9+ free parameters and fails to move meaningfully; concatenating all 5 assets’ residuals (55 total) restores a well-posed system.
2. **Compute budget.** $N_{\text{CAL_PATHS}} = 10$, $T_{\text{CAL}} = 1342$ days, 24 p -grid points $\times$ 10 paths = 240 simulations per hyperparameter evaluation, ≈ 20 –40s each; run in checkpointed chunks (each evaluation’s result saved immediately, since one evaluation alone can approach a single compute session’s time budget), capped at 200 evaluations.
3. **A checkpoint bug was found and fixed during development.** An earlier chunked-restart implementation re-seeded its own per-evaluation random-number counter to 0 at the

Algorithm 1 BOUNDEDLM_HYPERPARAMETERS (outer) calling GRIDAVERAGESMOOTH_FULLP (inner)

Require: θ_0 (A2-981003 hyperparameters + N_r, N_z); bounds; **max_nfev**

Ensure: best hyperparameter combination and its residual sum of squares

```

1: function RESIDUALS( $\theta$ )
2:   decode  $\theta \rightarrow (7 \text{ hyperparameters}, N_r, N_z)$ 
3:    $\overline{st} \leftarrow \text{GRIDAVERAGESMOOTH\_FULLP}(\text{hyperparameters})$  ▷ Algorithm 2
4:   for each of the 5 market assets, each  $k = 1, \dots, 11$  do
5:      $r_k^{(\text{asset})} \leftarrow \sqrt{w_k} (\overline{st}_k - c_k^{(\text{asset})}) / s_k^{(\text{asset})}$ 
6:   end for
7:   return the concatenated 55-entry residual vector
8: end function
9:  $\theta^* \leftarrow \text{LEAST\_SQUARES}(\text{RESIDUALS}, \theta_0, \text{bounds}, \text{method}=\text{"trf"}, \text{diff\_step}=0.1)$ 
10: return  $\theta^*, \sum(\text{RESIDUALS}(\theta^*))^2$ 

```

Algorithm 2 GRIDAVERAGESMOOTH_FULLP(hyperparameters) – average smooth stylised facts over all 12 gridded entries of p

Require: fixed hyperparameters; 24-point randomised 2-level design over all 12 entries of p (wide bounds)

Ensure: averaged stylised-fact vector $\overline{st}$

```

1: stat_vectors  $\leftarrow []$ 
2: for each of the 24 grid points do
3:   build reservoir matrices from (hyperparameters, this grid point's full  $p$ )
4:   if inadmissible ( $\kappa_0 \geq 0$ ) then skip this grid point
5:   end if
6:   for  $q = 1, \dots, 10$  do simulate a path; compute its 11 SMOOTH stylised facts
7:   end for
8:   average those 10 paths' stylised facts  $\rightarrow$  one stat vector; append to stat_vectors
9: end for
10: return  $\overline{st} \leftarrow$  average of every vector in stat_vectors, statistic by statistic

```

start of every new process, so each new chunk's finite-difference Jacobian probe used the *exact same* simulation seeds as the previous chunk – silently re-evaluating identical points with identical noise instead of exploring further. Fixed by offsetting the seed counter by the number of evaluations already recorded in the checkpoint.

3 Results

3.1 Convergence

S_{avg} falls from 2.90 (start) to a best of 2.62 (evaluation 192) over the 200 iterations. **This 2.62 is a single noisy Monte-Carlo evaluation, not a repeated estimate of any architecture's true expected score, and should not be read as this architecture's typical performance.** Decoding the parameter vector at evaluation 192 gives exactly $N_r = 22, N_z = 29$ – the same architecture reported in §3.2 below – but that architecture's *repeated*-evaluation performance is 2.71–2.73 (§3.2, §3.3), not 2.62. The gap is the same running-minimum-of-a-noisy-quantity effect

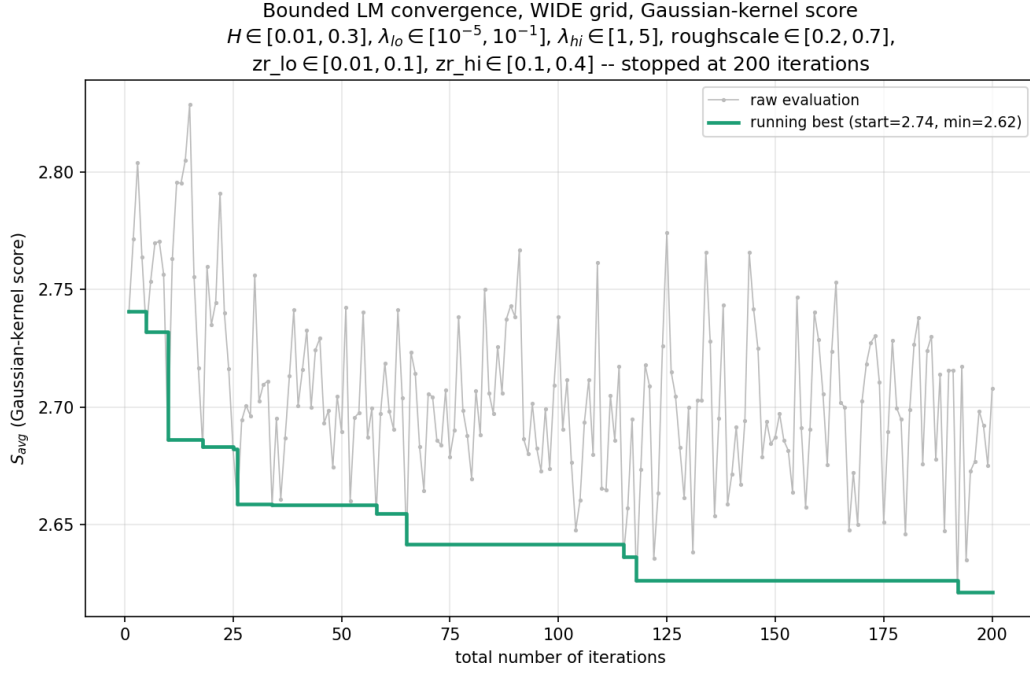


Figure 1: Bounded LM convergence, Gaussian-kernel score S_{avg} , capped at 200 iterations.

documented elsewhere in this model family: with $N_{CAL_PATHS} = 10$ paths per evaluation and 200+ evaluations, the luckiest single draw sits meaningfully below the architecture’s true average by chance alone. **The 2.71–2.73 figures are the reliable ones**; Figure 1’s running-best line should be read as an optimistic bound, not an achieved result.

3.2 Found architecture and stability

Bounded LM’s best point converges to $N_r = 22, N_z = 29$. A 5-seed repeat at this architecture (7 hyperparameters fixed) gives $S_{avg} = 2.725 \pm 0.043$ – **this, not Figure 1’s single-evaluation minimum of 2.62, is this note’s best estimate of the architecture’s true score.**

Table 1: Optimal hyperparameters (bounded LM argmin, all 12 entries of p gridded).

Hyperparameter	Start (A2-981003)	Found
N_r	48	22
N_z	20	29

3.3 (N_r, N_z) heat map

The heat map’s best cell, $N_r = 22, N_z = 40$, scores 2.708 – matching bounded LM’s own found point closely and confirming $N_r = 22$ as a genuinely good choice on this scale.

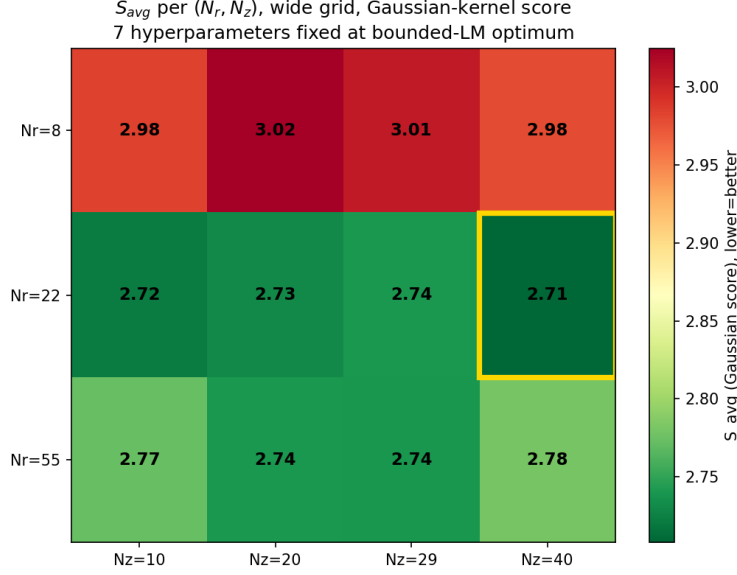


Figure 2: S_{avg} (Gaussian-kernel score) per (N_r, N_z) cell, 7 hyperparameters fixed at the bounded-LM optimum. Gold border = best cell found here.

3.4 Every stylised fact: market data vs. the optimal ESN architecture

Table 2 reports the complete 11-term stylised-fact vector at the optimal architecture ($N_r = 22, N_z = 29$, Table 1), averaged over the 24-point wide p -grid exactly as in §1.6, alongside each of the 5 real assets’ own targets.

Table 2: All 11 (smooth) stylised facts: each asset’s target vs. the optimal architecture’s, averaged over the full 12-parameter wide grid.

Stylised fact (w_k)	GPC	QCOM	INTC	LUV	NI	Optimal ESN
$\hat{H}$ (2.2)	0.0467	0.0529	0.0558	0.0459	0.0543	0.0538
mean ann. vol (1.4)	0.1847	0.3204	0.2696	0.2864	0.1780	0.2224
99.5% quantile ann. vol (0.6)	0.6961	1.1511	0.8996	1.0410	0.7150	0.7904
max ann. vol (0.2)	1.0316	1.7909	1.4867	1.4971	1.1223	1.2324
mean vol ACF (0.8)	0.2151	0.1864	0.2356	0.1962	0.2256	0.0976
Taylor gap (1.0)	0.0382	0.0474	0.0478	0.0367	0.0464	0.0127
Taylor fraction (0.8)	0.8387	0.8910	0.9544	0.8488	0.8693	0.6463
Zumbach asymmetry (1.0)	0.1081	0.1249	0.1296	0.1067	0.1027	0.0850
leverage corr. (1.1)	−0.0343	−0.0308	−0.0357	−0.0319	−0.0403	−0.0884
excess kurtosis (0.7)	5.443	4.473	4.563	3.661	4.985	3.967
max return ACF (0.5)	0.0948	0.0770	0.0948	0.0794	0.0814	0.0655

$\hat{H}$, mean annualised volatility, and both tail-volatility terms (99.5% quantile, max) land close to every asset’s own range – the architecture’s overall volatility level and persistence are well matched. Mean vol ACF, Taylor gap, Taylor fraction, and Zumbach asymmetry are all systematically smaller in magnitude than every asset’s target, and excess kurtosis under-shoots every target but LUV: the ESN’s simulated volatility clustering and leverage/asymmetry effects are directionally correct (the signs all match) but consistently weaker

than what the real data shows. **Leverage correlation is the one term where the ESN over-shoots:** -0.088 against a target range of -0.030 to -0.040 , more than double every asset’s own magnitude. These are the clearest remaining targets for improvement, and are consistent with this architecture family’s persistent, previously-documented difficulty fitting the volatility-clustering and asymmetry-related terms simultaneously alongside the volatility-level terms.

4 Limitations and Recommended Next Steps

1. **The 24-point fractional design was not checked for balance or adequacy.** A formal resolution-IV (or better) fractional factorial, or a larger design ($N = 48\text{--}64$), would give a more reliable average and let individual parameters’ effects be estimated rather than only their aggregate.
2. **The search was capped at 200 evaluations; whether it has fully converged is not certain.** Running further, or from multiple starting points, was not done here.
3. **Only one bounded-LM run, from one starting point, was performed.**
4. **Mean vol ACF, Taylor gap/fraction, Zumbach asymmetry, leverage correlation, and excess kurtosis (§3.4) remain the clearest targets for improvement.** All 5 are systematically off in the same direction (weaker clustering/asymmetry than every asset, except leverage which over-shoots) – whether this is fixable by further hyperparameter search, or requires freeing more of p specifically for these terms, was not tested here.